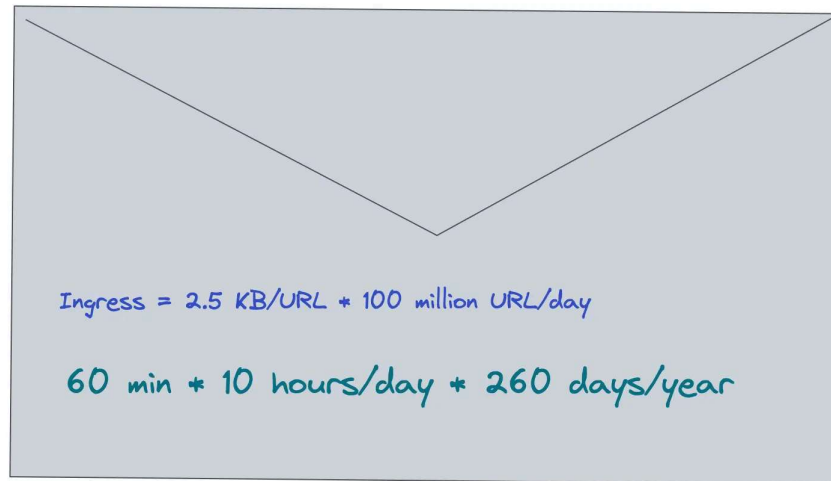


Back of the Envelope

Capacity Planning System Design

👤 NK included in 📁 Fundamentals

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- *Tech workers*
- *Students*
- *Engineering managers*

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What is a Back of the Envelope Calculation?

The back of the envelope are estimates generated using a combination of [thought experiments](#) and performance numbers through simple arithmetic. The performance numbers of the services involved should be known to conduct back of the envelope calculation. The unknown performance numbers can be determined through prototyping. For instance, measuring the write performance of the cache server can be determined through a prototype ¹.

The back of the envelope calculation is commonly used to evaluate different architectural designs. Although the optimal design depends on the system requirements, the back of the envelope calculation provides an opportunity to compare different architectural designs without prototyping. In layman's terms, back of the envelope analysis is a framework for comparing different variants of architectural design ¹.

The back of the envelope calculation is also known as **capacity planning**. In general, measuring system performance and performing back of the envelope is important in any deployment to satisfy the service level agreement (**SLA**) ². The following are some of the popular reasons to perform back of the envelope calculations:

- decide whether the database should be partitioned

- identify potential bottlenecks in the system

Understanding a Back of the Envelope Calculation

The back of the envelope calculation should be performed for efficient resource management and to meet scalability requirements ³. The different approaches to scaling a web service are the following:

- vertical scaling
- horizontal scaling
- diagonal scaling

Vertical scaling is achieved by increasing the number of CPU cores, memory, and disk storage on a single server. However, the server will reach a point of saturation where increasing the number of components does not further increase the throughput.

Horizontal scaling is the method of adding multiple machines to scale the web service. The drawback of horizontal scaling is increased complexity.

Diagonal scaling is a combination of both vertical and horizontal scaling. Diagonal scaling brings the best of both worlds by increasing or upgrading the components on a single server and replicating the server. Diagonal scaling offers relatively better performance and a reasonable price.

The back of the envelope calculation allows you to identify the sweet spot between diagonal scaling and a feasible price point. The following concepts should be well understood to perform a quick back of the envelope calculation:

- powers of two
- availability numbers
- latency numbers every programmer should know

The power of two is a handy reference to perform the back of the envelope.

Power	Exact Value	Approx Value	Bytes
7	128		
8	256		
10	1024	1 thousand	1 KB
16	65,536		64 KB
20	1,048,576	1 million	1 MB
30	1,073,741,824	1 billion	1 GB
32	4,294,967,296		4 GB

A byte is a unit of digital information that consists of 8 bits. The data size of the most commonly used data types can be helpful for the back of the envelope ^{4, 5}.

Data Type	Size (byte)
int	4
float	8
boolean	4
UTF-8 character	1
UTF-8 in Chinese	3
UNIX timestamp	4

|

Availability Numbers

High availability is the ability of a service to remain reachable and not lose data even when a failure occurs. High availability is typically measured in terms of a percentage of uptime over a given period. The uptime is the

Availability (percentage)	Downtime per year
99	3.6 days
99.99	52 minutes
99.999	5 minutes
99.9999	31 seconds

In addition to measuring uptime, high availability can also be measured by other metrics such as mean time between failures (**MTBF**), which measures the average time between system failures, and mean time to repair (**MTTR**), which measures the average time it takes to restore a system after a failure.

A service-level agreement (**SLA**) is an explicit or implicit contract between a service provider and the users. The SLA documents the set of services the service provider will offer to the user and defines the service standards the provider is obligated to fulfill.

| Latency Numbers Every Programmer Should Know

The [interactive latency tool](#) can be used to visualize the latency of components and operations. The numbers shown are supposed to give a rough idea of the latency of different operations ^{6, 4, 7}.

Operation	Time
L1 cache reference	0.5 ns
Branch mispredict	5 ns
L2 cache reference	7 ns
Mutex lock/unlock	25 ns
Main memory reference	100 ns

Compress 1K bytes with Zippy	10 us
Send 1 KB bytes over 1 Gbps network	10 us
Read 4 KB randomly from SSD*	150 us
Read 1 MB sequentially from memory	250 us
Round trip within same datacenter	500 us
Read 1 MB sequentially from SSD*	1 ms
HDD seek	10 ms
Read 1 MB sequentially from 1 Gbps	10 ms
Read 1 MB sequentially from HDD	30 ms
Send packet CA->Netherlands->CA	150 ms

Useful Conversions

Time	Time (seconds)
1 ns	10 ⁻⁹ seconds
1 us	10 ⁻⁶ seconds
1 ms	10 ⁻³ seconds

The latency numbers can be summarized as the following ¹:

- memory is fast and disks are slow
- network bandwidth can be saved by using a compression algorithm
- writes are more expensive than reads
- globally shared data is expensive
- at most 7 round trips between inter-region data centers per second are possible
- approximately 2000 round trips per second can be achieved within a data center

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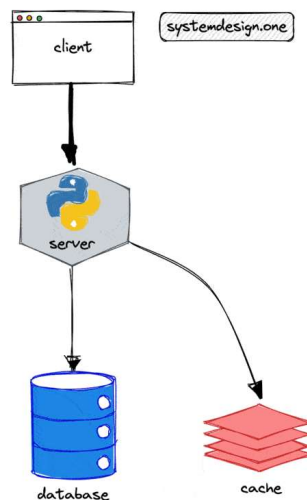
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Back of the Envelope Calculation in System Design Interview

The back of the envelope calculation is not mandatory during a system design interview. You should discuss with your interviewer whether back of the envelope calculations are required. The back of the envelope calculations should not take more than five minutes of your system design interview unless stated by the interviewer.

You can visit similarweb.com to determine the approximate traffic of popular websites across the internet. It is reasonable to ask the interviewer for the Query per second (**QPS**) during requirements clarification to make the back of the envelope calculation easier.



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The back of the envelope analysis is supposed to be approximate and is based on assumptions. You are expected to make informed decisions and discuss trade-offs based on the back of the envelope calculations.

| Tips for Back of the Envelope Calculations

The back of the envelope calculation is about evaluating your problem-solving skills in the system design interview. The following tips can be helpful for back of the envelope calculation ^{5, 8, 9}:

- rounding and approximation of numbers (For example, 83 becomes 100)
- label the units for easy calculation
- always convert to operations per second
- 1 million requests/day = 12 requests/second
- explicitly specify the assumptions
- use numbers along with an abbreviation for the magnitude (1000 = 1K)

You should make reasonable assumptions, ask clarifying questions and communicate your thought process with the interviewer to achieve success in the system design interview.

| Why Is the Back of the Envelope Analysis Important?

The back of the envelope analysis is important due to the following reasons ^{6, 1, 8}:

- evaluating different architectural designs
- validating the proposed architectural design
- uncovering bottlenecks
- ensuring high availability
- improving cost efficiencies

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spikes

- predicting system capacity requirements

| Types of Back of the Envelope Calculations in System Design Interviews

The most common types of back of the envelope calculations in system design interviews are the following:

- traffic estimation
- storage estimation
- memory estimation
- bandwidth estimation
- resource estimation
- latency estimation

| Latency Estimation

When an API endpoint should merge data from multiple resources and the data fetching process is sequential, the latency is calculated as follows:

💡 Latency for sequential data fetch

$$\text{latency} = \text{latency_resource_1} + \text{latency_resource_2}$$

When an API endpoint should merge data from multiple resources and the data fetching process is parallel, the latency is calculated as follows:

💡 Latency for parallel data fetch

$$\text{latency} = \max(\text{latency_resource_1}, \text{latency_resource_2})$$

| Resource Estimation

You can assume that a modern server contains 32 cores and the system latency should be less than 250 ms for a [real-](#)

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of the envelope calculation. The following formula is used to calculate the Query per second (**QPS**) given the number of CPU cores and CPU time per second ¹⁰:

💡 Query per second

QPS = number of CPU cores / average time for a request in seconds

The initial design should focus on supporting the average traffic because it is the usual condition. The peak traffic must also be considered to make the system fault tolerant and highly available. The different approaches to determining peak traffic are the following ⁵:

- 10% per hour rule
- peak traffic = 2 * average traffic

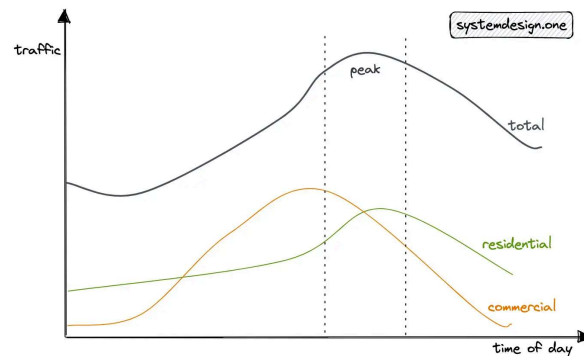


Figure 2: Real-world peak traffic

With the 10% per hour rule, when daily traffic is 1 million, you can assume that 10% of the daily traffic occurs within 1 hour. This would result in approximately 30 QPS. Alternatively, you should discuss with your interviewer whether it's fine to assume the peak traffic to be twice the average traffic.

Back of the Envelope Calculation Example

Traffic

The [URL shortener](#) is a read-heavy service. The Daily Active Users (**DAU**) for writes is 100 million. The Query Per Second (**QPS**) of reads is approximately 100 thousand.

Description	Value
DAU (write)	100 million
QPS (Write)	1000
read: write	100: 1
QPS (read)	100 thousand
time of persistence for a URL	5 years

Storage

The shortened URL persists by default for 5 years in the data store. Each character size is assumed to be 1 byte. A URL record is approximately 2.5 KB in size. Set the replication factor of the storage to a value of at least three for improved durability and disaster recovery.

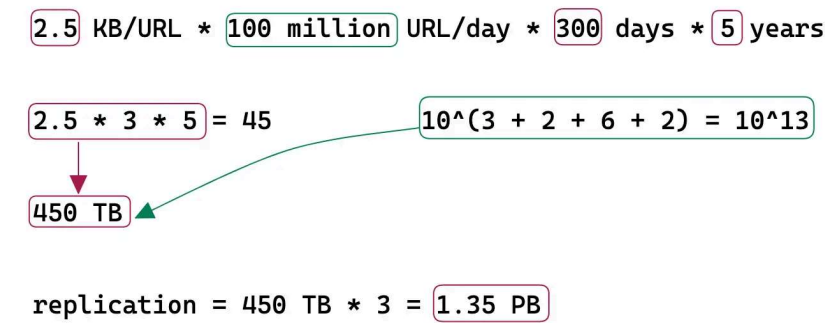


Figure 3: Back of the envelope; Storage

Bandwidth

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$$2.5 \text{ KB/URL} * 100 \text{ million URL/day} * 10^{(-5)} \text{ day/sec}$$

$$2.5 \text{ MB/sec}$$

$$10^{(3 + 2 + 6 - 5)} = 10^6$$

Figure 4: Back of the envelope; Ingress

Egress is the network traffic that exits the servers (server responses).

$$2.5 \text{ KB/URL} * 10 \text{ billion URL/day} * 10^{(-5)} \text{ day/sec}$$

$$250 \text{ MB/sec}$$

$$10^{(3 + 10 - 5)} = 10^7$$

Figure 5: Back of the envelope; Egress

Memory

The URL redirection traffic (egress) is cached to improve the latency. Following the [80/20 rule](#), 80% of egress is served by 20% of URL data stored on the cache servers. The remaining 20% of the egress is served by the data store to improve the latency. A Time-to-live (TTL) of 1 day is reasonable.

$$10 \text{ billion URL/day} * 2.5 \text{ KB/URL} * 0.2$$

$$5 \text{ TB/day}$$

$$10^{(10 + 3 - 1)} = 10^{12}$$

Figure 6: Back of the envelope; Memory

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[Fermi estimation](#) is a method for arriving at an estimate of an uncertain variable of interest. More architectural designs can be evaluated when cheap and quick reasonable estimates of the numbers can be made ⁹.

The performance of a system is not directly proportional to the system's capacity. In layman's terms, the performance of the system depends on the asymptotic complexity of the system implementation. The service should be monitored and benchmarked to perform back of the envelope analysis. For example, it is necessary to know whether the load is read or write heavy, and the rate of concurrency to [scale](#) a database ¹¹.

If you are hosted on the cloud, the service tiers 0 and 1 should be used for the production load to attain high performance. Service tiers 2 and 3 should be used for the testing load to reduce costs. The system can be configured to switch to a different instance type when the current instance type capacity is fully utilized ¹².

The historical metrics are required to make an educated forecast. The system-level and application-level statistics make up the historical metrics. The estimation provided by the marketing or product team should also be combined with the historical metrics to determine a robust forecast ³.

The linear regression model for forecasting the traffic will not capture the seasonality of the time series. However, in the absence of seasonality, the linear regression model is preferred as it is computationally less expensive than other models. The [ARIMA models](#) can be used to explicitly model trend and seasonality in time series for robust forecasts ¹³. The system should degrade gracefully when the capacity limits are exceeded. The following methods can be used for graceful degradation of the service ³:

- disable heavy load features
- cache dynamic content into static bits
- serve stale content

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load. On the other hand, the time and effort to run a simulation are wasted because the application logic evolves quickly. On top of that, you should not over-provision the servers to handle peak traffic to save costs and reduce operational complexity ³.

| What Does Real World Back of the Envelope Calculation Involve?

The real-world back of the envelope calculation involves the following ¹⁴:

1. provision performance environment
2. determine the single host load pattern
3. develop and run load tests
4. analyze and extrapolate results

The tools for back of the envelope calculation are the following ¹⁵:

- load and capacity testing
- stress and volume testing

[Locust](#) is a popular tool for load testing. The perfect parity with production makes a good load test. The production data can be scrubbed to synthetically generate the production load. In addition, the following metrics should be monitored during a load test ¹⁴:

- client-side metrics (error rate, latency)
 - server-side metrics (response time, CPU, memory, thread counts)
 - auto-scaling trigger frequency
-
-
-

Summary

The back of the envelope calculation is a vital skill for software engineers preparing for the system design interview. By taking a proactive approach through the back